

Modular Modeling Documentation Structure

Directory Organization

Status

-  Completed: 2/12
-  Under review: 3/12
-  To be developed: 6/12
-  Deprecated: 1/12

projectsPortfolioModel/

Portfolio composition and project-level characteristics

1.  [profitMarginsComposistions.md](#) - Profit margin distributions by category
2.  [projectCounts&BACsDistributions.md](#) - Portfolio size and BAC distributions
3.  [projectSCurves.md](#) - S-curve cashflow model (Beta distribution)
4.  (deprecated - merged into the projectSCurves) [projectDurations.md](#) - Duration distributions and start time staggering
5.  [projectRevenuePlans.md](#) - Revenue recognition and payment structures

uncertaintyModel/

Stochastic performance and uncertainty modeling

6.  [projectsPerformances.md](#) - Contractor performance uncertainty (SPI/CPI)
7.  [projectsRevenues.md](#) - Payment delays and revenue uncertainty
8.  [portfolioTemporalStructre.md](#) - Temporal structure and seasonal effects

instanceGeneratorModel/

Synthetic data generation and validation

9.  [portfolioGenerator.md](#) - Complete instance generation algorithm
10.  [datasetValidation.md](#) - Validation framework and sanity checks

RLModel/

Reinforcement learning framework

11.  [rewardFunction.md](#) - Reward function design and components
12.  [MDP.md](#) - MDP formulation (state, action, transition, reward)

Template

All documents follow the structure in `0. promptTemplate.md`